

Triqubit SU(3) Encoding: Exact Error-Detection and QEC No-Go Status

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Abstract

The one-hot color carrier $\mathcal{H}_{\text{color}} = \text{span}\{|100\rangle, |010\rangle, |001\rangle\}$ is an exact three-qubit realization of the fundamental $\mathbb{C}^3$ carrier of SU(3). We prove the precise noise statement: every individual single-qubit X_i or Y_i error is detectable as leakage from the one-hot sector, while single-qubit phase errors Z_i are not detected by the occupation projector. More strongly, the Knill–Laflamme conditions fail for the error set $\{I, X_1, X_2, X_3\}$, so the one-hot sector cannot correct an unknown single bit flip. It is therefore a leakage-detecting qutrit embedding, not a quantum error-correcting code and not evidence for a simulation ontology.

1 Definitions

Let

$$\mathcal{H}_{\text{color}} = \text{span}\{|100\rangle, |010\rangle, |001\rangle\} \subset (\mathbb{C}^2)^{\otimes 3},$$

and let

$$P_{\text{color}} = |100\rangle\langle 100| + |010\rangle\langle 010| + |001\rangle\langle 001|$$

be the orthogonal projector. The Pauli operator acting on qubit i is denoted by X_i, Y_i, Z_i .

An error E is detectable by the code projector precisely when

$$P_{\text{color}} E P_{\text{color}} = c_E P_{\text{color}}$$

for a scalar c_E . A finite error set $\{E_a\}$ is correctable only if the Knill–Laflamme conditions hold:

$$P_{\text{color}} E_a^\dagger E_b P_{\text{color}} = c_{ab} P_{\text{color}} \quad \text{for all } a, b.$$

2 Exact status theorem

Theorem 2.1 (Leakage detection and correction no-go). *For the one-hot color sector the following statements hold.*

1. For each $i = 1, 2, 3$,

$$P_{\text{color}} X_i P_{\text{color}} = 0, \quad P_{\text{color}} Y_i P_{\text{color}} = 0.$$

Hence every individual X_i or Y_i error, and every coherent linear combination of such operators, is detectable by the two-outcome measurement $\{P_{\text{color}}, I - P_{\text{color}}\}$ as leakage.

2. For each i , $P_{\text{color}}Z_iP_{\text{color}}$ is not proportional to P_{color} . Occupation-number measurement therefore does not detect a general single-qubit phase error.
3. The error set $\{I, X_1, X_2, X_3\}$ is not correctable. For example,

$$P_{\text{color}}X_1X_2P_{\text{color}} = |100\rangle\langle 010| + |010\rangle\langle 100|,$$

which is not a scalar multiple of P_{color} . Thus the Knill–Laflamme condition fails.

4. The three-dimensional sector is not a standard qubit Pauli-stabilizer code, whose code-space dimension is a power of two.

Proof. A one-hot basis state has Hamming weight one. Acting with X_i changes that weight to zero or two; $Y_i = iX_iZ_i$ has the same bit-flip support. Their images are therefore orthogonal to $\mathcal{H}_{\text{color}}$, proving the first identities.

On the ordered color basis ($|100\rangle, |010\rangle, |001\rangle$), each compressed Z_i has one sign opposite to the other two. It is diagonal but not scalar, proving non-detectability by P_{color} .

For correction, choose $E_a = X_1$ and $E_b = X_2$. Since Pauli X is Hermitian, $E_a^\dagger E_b = X_1X_2$, whose compression exchanges the red and green basis states and annihilates the blue basis state. It is not scalar, so the necessary and sufficient Knill–Laflamme condition fails. The stabilizer statement follows from the standard dimension formula 2^k for a qubit Pauli-stabilizer code space. $\square$

Claim ledger

- **GAP-SU3-TRIUBIT-LEAKAGE: CLOSED [L1].** Single-qubit X_i and Y_i errors are exactly detected as leakage.
- **GAP-SU3-TRIUBIT-QEC: CLOSED AS NO-GO [L1].** The present one-hot encoding does not correct an unknown single X_i error and does not detect general Z_i errors.
- No decoder, recovery channel, physical penalty Hamiltonian, or UBT-derived noise model is supplied by this theorem.

Corollary 2.2 (No simulation inference). *The representation-level code-subspace structure does not imply that quarks are an error-correcting code of an underlying simulator, nor that physical reality is a simulation. Such an ontological interpretation is logically independent and remains speculative.*